

FACE-TRACKING SYSTEM

Using A Laptop, ESP32, And YOLOE

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The system combines a laptop running the YOLOE (You Only Look Once – Everything) face-detection model with an ESP32 for servo control and sensor interfacing. This division of tasks enables face tracking at more than 50 frames per second (FPS), with the laptop handling computationally intensive AI inference and the ESP32 managing real-time hardware control and sensing.

This hybrid design combines the processing power of a laptop with the low-latency hardware control capabilities of the ESP32. Conventional implementations that run the detection model entirely on a Raspberry Pi 5, even with a Hailo-8L AI

accelerator, typically achieve around 30 FPS, while software-only implementations generally operate at just

3-5 FPS and require a GUI, which further increases the processing load. In comparison, the same YOLOE or YOLOv8-Face models running on a mid-range laptop can consistently achieve 50+ FPS, with even higher frame rates when GPU acceleration is available.

While a laptop excels at AI inference, it is not well suited to direct, low-latency control of servos and sensors. To overcome

this limitation, the ESP32 controls the pan-tilt servos, interfaces with a VL53L0X time-of-flight (ToF) distance sensor, and drives a 128 × 64 OLED display. The laptop and ESP32 communicate using the lightweight ESPConnect protocol, enabling the laptop to transmit face coordinates, servo commands, and face-count information, while the ESP32 returns distance measurements and performs real-time hardware control.

The system employs a closed-loop raster-scan algorithm. During idle operation, the camera performs rapid horizontal sweeps from 0° to 180°, incrementally adjusting the tilt angle before scanning back in the opposite direction. Once one or more faces are detected, the camera continuously tracks the centroid of the detected group instead of locking onto a single face. This behaviour resembles the auto-tracking used in professional video-conferencing cameras, helping keep all detected subjects within the camera's field of view. An external USB webcam is mounted directly

BILL OF MATERIALS

Components	Description	Quantity
ESP32 development board	Wi-Fi-enabled microcontroller for hardware control	1
PCA9685 PWM driver	16-channel PWM servo driver board	1
Hobby servo motors	Standard 180° pan and tilt servos (expandable)	2
VL53L0X ToF sensor	Time-of-flight distance sensor	1
128×64 OLED display	SH1106 or SSD1306 I ² C OLED module	1
USB webcam	External camera mounted on pan-tilt assembly	1
5V regulator	Voltage regulator for servo power rail	1
100μF capacitor	Smoothing capacitor across 5V power rail	1
Protection diode	Reverse-polarity protection diode	1
Power supply	7V-9V DC source for servo power rail	1
Connecting wires	Assorted jumper wires and cables	As required

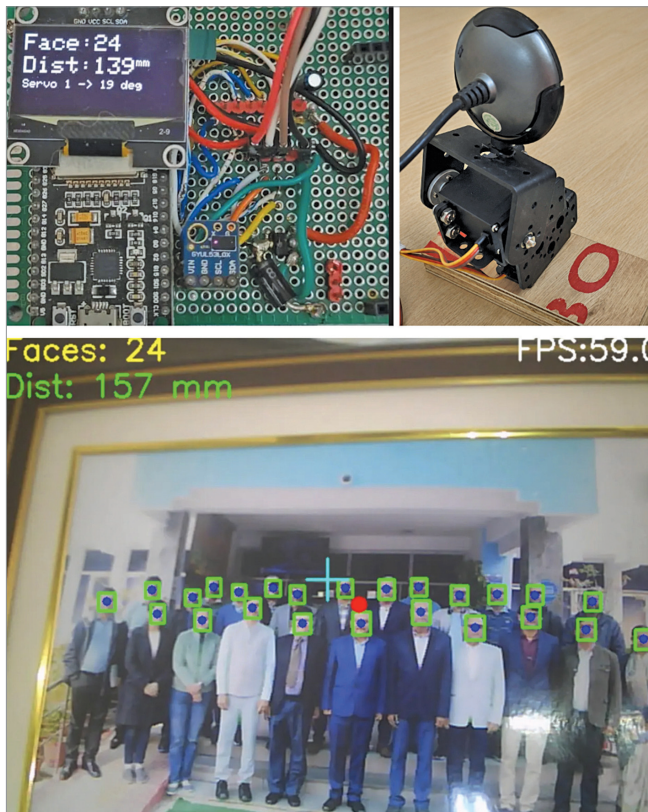


Fig. 1: Authors' prototype and testing setup